

Credit Risk Prediction

Exploratory Data Analysis & Logistic Regression Model

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Tools & Technologies

Python · Pandas · Scikit-learn · Seaborn · Matplotlib · Logistic Regression

Dataset: Credit Risk Dataset (Kaggle) · 32,581 loan records

Executive Summary

This project applies data science techniques to predict credit default risk using a dataset of 32,581 loan records. Through exploratory data analysis, customer segmentation, and logistic regression modeling, the project identifies the key financial and demographic attributes that drive loan default probability. The final model achieves 84% accuracy and an ROC-AUC score of 0.853 — providing a practical framework for automated credit decision making.

Business Problem

Financial institutions face significant losses from loan defaults every year. The challenge is identifying which borrowers are likely to default before approving their applications. Traditional manual review processes are slow, inconsistent, and unable to scale. A data-driven credit risk model enables banks to:

- Automatically approve low-risk applications — improving customer experience and processing speed
- Flag medium-risk applications for human review — focusing analyst time where it matters most
- Automatically reject high-risk applications — protecting the institution from losses
- Quantify expected losses and maintain appropriate loan loss reserves

Primary Question: Which borrower and loan attributes most strongly predict the probability of default?

Secondary Question: How can a bank use predictive scoring to optimize its loan approval process?

Goal: Build a model that outputs a default probability score for each applicant and assigns them to a risk Tier.

Dataset Overview

The dataset contains 32,581 loan records with 12 features covering borrower demographics, loan characteristics, and credit history. The target variable is `loan_status` — 0 indicating the loan was repaid and 1 indicating default.

Feature	Type	Description
person_age	Numeric	Borrower age
person_income	Numeric	Annual income
person_home_ownership	Categorical	Rent / Own / Mortgage
person_emp_length	Numeric	Years employed (895 missing — filled with median)
loan_intent	Categorical	Purpose of loan
loan_grade	Categorical	Bank risk rating A through G
loan_amnt	Numeric	Loan amount requested
loan_int_rate	Numeric	Interest rate charged (3,116 missing — filled with median)
loan_status	Binary	TARGET: 0 = Paid back, 1 = Defaulted
loan_percent_income	Numeric	Loan amount as % of annual income
cb_person_default_on_file	Categorical	Has borrower defaulted before? Y/N
cb_person_cred_hist_length	Numeric	Years of credit history

Methodology

This project follows the CRISP-DM framework — the industry standard for data science projects:

1. **Business Understanding:** Define the credit risk problem and success criteria
 2. **Data Understanding:** Explore distributions, identify missing values, understand each feature
 3. **Data Preparation:** Fill missing values with median imputation, encode categorical variables
 4. **Modeling:** Train logistic regression model with 80/20 train/test split
 5. **Evaluation:** Assess accuracy, ROC-AUC, precision, recall, and confusion matrix
 6. **Recommendations:** Translate findings into actionable business decisions
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Key Findings from Exploratory Data Analysis

Finding 1 — Default Rate Distribution

The dataset shows a 78.2% repayment rate and 21.8% default rate — significantly higher than the typical industry average of 5-10%. This elevated default rate suggests the portfolio carries above-average credit risk and underscores the need for robust predictive modeling.

Finding 2 — Loan Grade is the Strongest Categorical Predictor

The bank's internal loan grading system proves highly predictive. There is a critical risk threshold between Grade C and Grade D — below Grade C the default rate stays under 21%, but at Grade D it jumps to 59% and climbs to nearly 100% for Grade G borrowers.

Grade	Default Rate	Risk Level
A	10.0%	LOW
B	16.3%	LOW
C	20.7%	LOW-MEDIUM
D	59.0%	HIGH
E	64.4%	HIGH
F	70.5%	VERY HIGH
G	98.4%	EXTREME

Finding 3 — Loan Intent Impacts Default Risk Debt consolidation loans carry the highest default risk at 28.6% — nearly double the rate of venture loans at 14.8%. Borrowers taking loans to cover existing debt are already in financial distress, making them significantly more likely to default.

Finding 4 — Income is a Critical Risk Factor

Low-income borrowers (under \$30k annually) default at 45.5% — nearly five times the rate of borrowers earning \$100k-\$200k (9.2%). Notably, the \$200k+ segment shows slightly elevated risk compared to the \$100k-\$200k segment, suggesting very high earners take on proportionally larger debt obligations.

Finding 5 — Strongest Numeric Predictors (**Correlation Analysis**)

Variable	Correlation with Default	Interpretation
loan_percent_income	0.38 (positive)	Higher debt burden = more defaults
loan_int_rate	0.32 (positive)	Higher interest rate = more defaults
person_income	-0.14 (negative)	Higher income = fewer defaults
loan_amnt	0.11 (weak)	Loan size alone is not a strong predictor

Predictive Model — Logistic Regression

A logistic regression model was trained on 80% of the dataset (26,064 customers) and evaluated on the remaining 20% (6,517 customers). The model uses 22 features after one-hot encoding of categorical variables.

Metric	Score	Interpretation
Overall Accuracy	84%	Correctly predicted 84 out of 100 customers
ROC-AUC Score	0.853	Good predictive power (0.85+ is considered strong)
Precision (Default)	75%	When model flags a default, it is correct 75% of the time
Recall (Default) — 50% threshold	42%	Model catches 42% of actual defaulters
Recall (Default) — 30% threshold	69%	Improved to 69% after threshold optimization

Threshold Optimization — The \$3.95 Million Improvement

The default classification threshold was lowered from 50% to 30% to improve default detection. This single adjustment reduced missed defaults from 839 to 444 customers — saving approximately \$3.95 million in potential losses (assuming an average loan of \$10,000).

Metric	50% Threshold	30% Threshold	Improvement
Missed Defaults	839	444	-395 customers
Potential Loss	\$8,390,000	\$4,440,000	-\$3,950,000 saved
Default Recall	42%	69%	+27 percentage points

Customer Risk Tier Distribution

The model assigns every loan applicant to one of three risk tiers — enabling automated decision making at scale while focusing human review where it adds the most value.

Risk Tier	Threshold	Customers	Action
LOW RISK	< 30%	4,901 (75%)	Auto-Approve
MEDIUM RISK	30% - 60%	1,002 (15%)	Manual Review
HIGH RISK	> 60%	614 (9%)	Auto-Reject

Business Recommendations

1. Apply Stricter Underwriting for Grade D and Below

The sharp jump in default rate from Grade C (20.7%) to Grade D (59.0%) represents the most actionable finding in this analysis. Banks should require additional collateral, co-signers, or higher down payments for all Grade D and below applications before approval.

2. Restrict Debt Consolidation Lending

Debt consolidation loans default at 28.6% — the highest of any loan purpose. Banks should apply stricter debt-to-income requirements for these applications and consider capping loan amounts for consolidation Purposes.

3. Cap Loan-to-Income Ratio at 35% for Auto-Approval

loan_percent_income is the strongest numeric predictor of default (correlation 0.38). Implementing a hard cap of 35% loan-to-income ratio for automatic approval would significantly reduce default exposure while having minimal impact on creditworthy borrowers.

4. Use 30% Classification Threshold in Production

Lowering the classification threshold from 50% to 30% reduces missed defaults by 395 customers and saves approximately \$3.95 million per loan cohort. The trade-off of flagging more medium-risk customers for manual review is justified by the significant reduction in default losses.

5. Retrain Model Quarterly

Credit risk patterns change with economic conditions — interest rate changes, unemployment shifts, and housing market movements all affect borrower behaviour. The model should be retrained quarterly with new loan data to maintain predictive accuracy and capture evolving risk patterns.

Next Steps — Model Improvement

- **Implement XGBoost or Gradient Boosting** — industry standard for credit risk, expected ROC-AUC above 0.90
- Apply **SMOTE (Synthetic Minority Oversampling)** to address class imbalance and improve default recall
- **Add macroeconomic features** — unemployment rate, GDP growth, interest rate environment
- Build a real-time scoring API so every new loan application is scored automatically on submission
- **Conduct fairness audit** — ensure model does not discriminate by protected characteristics

Conclusion

This project demonstrates that machine learning can meaningfully improve credit risk assessment. The logistic regression model achieves 84% accuracy and an ROC-AUC of 0.853 — strong performance for a baseline model. Through threshold optimization alone, the model reduces potential losses by nearly \$4 million per loan cohort compared to the default 50% threshold. The key insight from this analysis is that loan grade, debt-to-income ratio, and interest rate are the most powerful predictors of default. Banks that implement data-driven scoring systems based on these variables can significantly reduce credit losses while maintaining a fair and efficient loan approval process. [Kaosisochukwu \(Kosi\) Nweke](#) · github.com/thiskosy · kosinweke.e@gmail.com · linkedin.com/in/kosy04